

* Advantages:-

- 1) Each and every stage of SDLC is tested.
- 2) The outputs or deliverable are given simultaneously or parallelly, because of which product gets built at a high speed.
- 3) Downward flow of defect are reduced.
- 4) Total investment is less.

* Drawbacks:-

- 1) It involves lot of documentation work.
- 2) Initial investment is high.

* Applications:-

- 1) When customer is expecting high quality product.
- 2) When customer wants complex or long term project.

* Agile Model:-

• SDLC

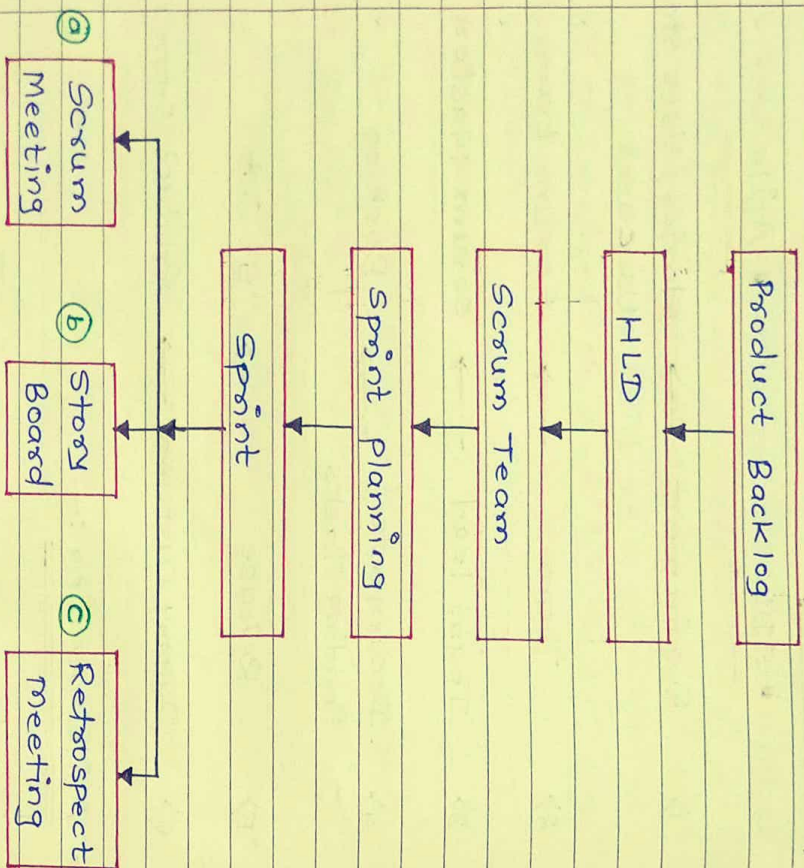
• Agile

- 1) Requirement → Stories / User stories / Use cases.
- 2) Team → Scrum team
- 3) Team lead → Scrum Master
- 4) Documents / Pending Tasks → Backlog
- 5) Release → Sprint
- 6) Dummy Customer → Product Owner.

* Agile meaning:-

- Flexible.
- Flexible in terms of handling changes from customers, changes like addition, removal and modification of features.
- It is an incremental process model.
- It is very realistic approach to software development.
- It is useful in all the sectors, where flexibility of software development is more.

* Flow chart of Agile Model :-



1) * Product Backlog and HLD :-

- High level story

Create A/c	Apply loan
Login	Apply OD
Logout	Apply check
Remove Ac	Apply DD
check Bal	claim
Add Bal.	Repay
Withdrawal	Apply Insu
Settings	etc

• Product Backlog

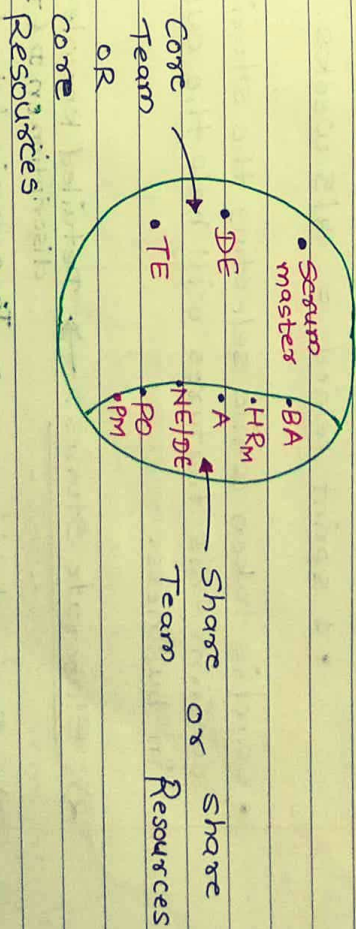
- By :-
Project Manager
OR
Product Owner.

- It is a document which contains high level stories.
- This ~~are~~ are usually prioritised list of stories.
- It is usually managed by Project manager or product owner.
- It is a high level story (document).

* Stake holders :- 1) Important people in project

- 3) Imp. decision makers.
- 3) e.g.:- PM, PO, BA, HRM, Architect, etc.

2) * Scrum Team :-



- A group of people come together to deliver committed increment or features, is "Scrum Team".
- Usually the team has two parts :-

- a) Core team -> Scrum master, DE, TE.
- b) Share team -> PM, PO, BA, HRM, A,

NE/DE
(Network or Development Engineers Admin)

- * Spent Planning :-

- It is a document which contains high level stories taken from product backlog.

- We usually select stories which can be developed and tested within one sprint.

- 1 Release period = 1-1.5 years
- 1 sprint period = 4/5 weeks.

- Carc is taken while selecting the stories to confirm this features will help the customer in business.

2) Elaborate Stories. → Detailed knowledge and description on D & T stories.

3) Estimate time. → Time required to do that task and sprint preparation,

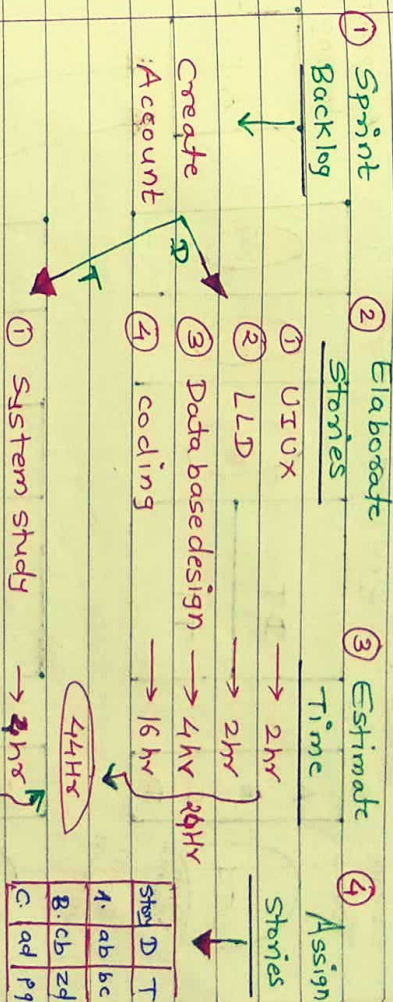
4) Assign stories → Stories given to TE & DE.
Assigning respective TE/DE.

5) Set acceptance criteria. → Acceptance limits set by PD.

- Sprint planning is generally done by the Scrum team (i.e. Scrum master, DE, TE) and product owner (PO).

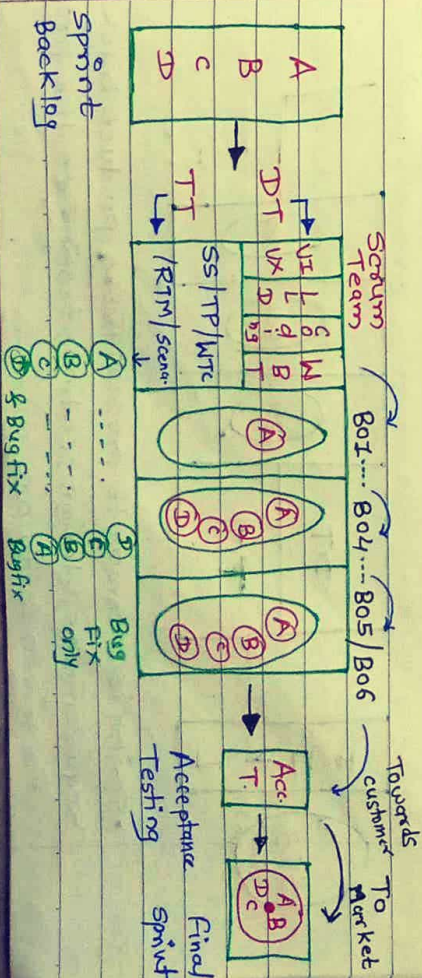
- Time required to complete one story (i.e. for DIT) is called "story point".

- Also, time required or estimated to complete one task, (i.e. for DIT the one story), is called "story point".



4) * Sprint :-

- sprint is nothing but an release.



* Average Time Periods :-

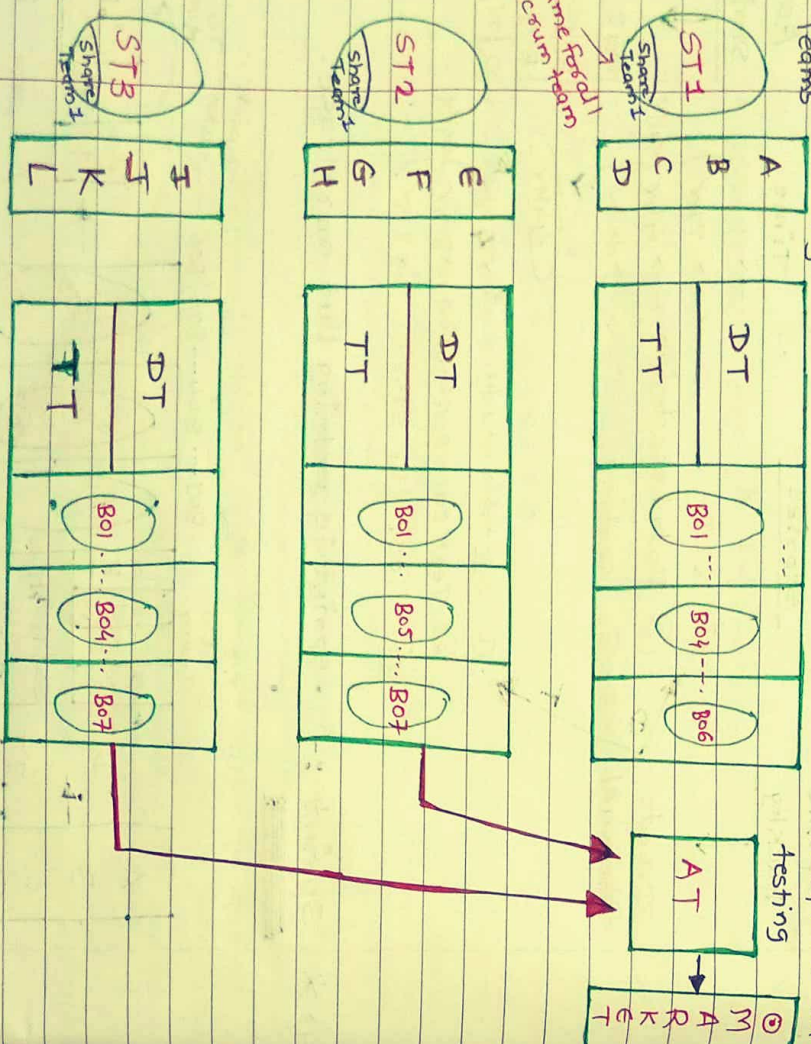
- 1) Sprint planning = 8 hr.
- 2) Sprint (1) = 4/5 weeks.
- 3) 1 Test cycle = 3/5 days
- 4) Test cycles in 1 sprint = 5-8 cycles
- 5) Scrum team members = 3-7 members.

Product Backlog

Scrum Teams

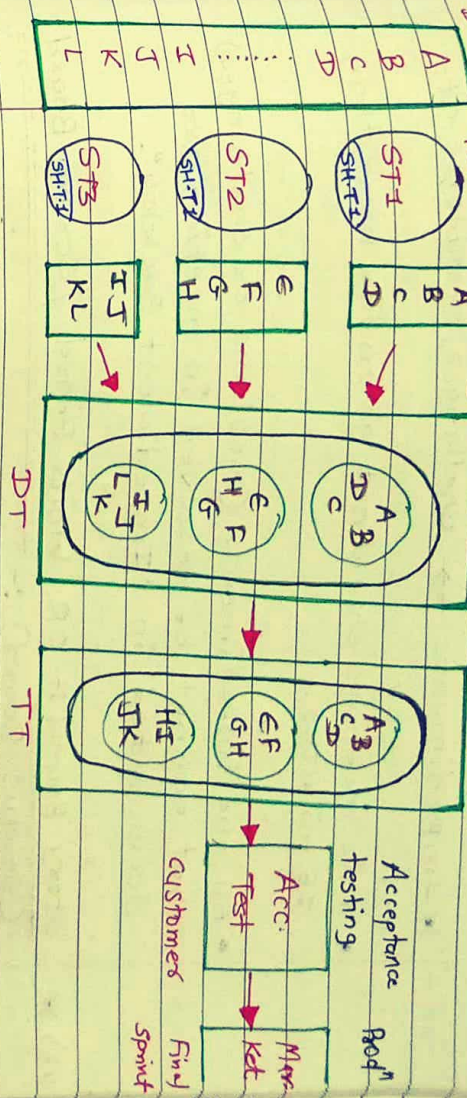
Dev and testing

* fig (a) :- Sprint having no of Scrum teams.



- Sprint is a complete process from product backlog or requirements, design, coding, testing to the production of a given software.

Product Backlog Scrum Teams Product Backlog Dev Environment Test Env.



* fig (b) :- Sprint having 3 Scrum Teams :-

- For getting larger numbers of stories in a one sprint, the Scrum team numbers should be increased.
- Also, for every Scrum team they have their Scrum masters and all Scrum team having only one share team or resources.
- e.g.:- In fig (a) and fig (b), there is stories are A → Z, But we consider important stories first and assigns a 3 Scrum teams with ABCD, EFGH & IJKL stories & backlogs develop and testing Purpose. Having same share team.

4a) * Scrum Meeting / Daily standup / Standup Meeting OR Daily Scrum :-

- It is a meeting to be conducted every day by Scrum masters with entire Scrum team.
- In this meeting, each engineer explains about tasks completed yesterday and challenges faced.
- The engineer will also explain the tasks to be completed on the same day & expected challenges.

- Impediments → challenges, problems, etc.
- This meetings should not go on for more than 15 minutes.
- All the impediments (problems or challenges) that cannot be resolved in meeting are documented in "Impediment Backlog".

4b) * Story Board :- OR White Board / Scrum Board (Status Board) :-

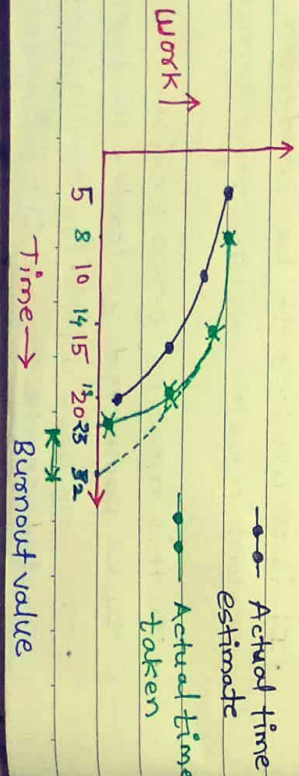
Test E	Pending Task	Completed Task	TE	To be done	doing	done
TE ₁	1/2/3/4		TE ₁	2/3	2	1
TE ₂	2/3/4	1	TE ₂	1/2/3		
TE ₃	4	1/2/3	TE ₃	3	3	1/2

• In General Technique

• Modern technique

- It is an representation of tasks that are given to a test engineer to complete in given period.
- It indicates the actual working efficiency of an engineer over the given tasks or stories.
- It gives Testers working stage at a particular day/time.

* Burnout chart OR Work Estimation chart :-



- Burnout chart is an graphical representation of work done to time taken for that work.
- Burnout Chart indicates or shows the how efficiently the work is done, which gives performance of the team.
- It show how fast you or team achieves the goals.
- Generally, this chart is always checked by Product owner.

* Chicken :-

- A engineer who is part of Scrum team, but not doing any work. The engineer either understanding process or learning, that person is called "chicken".
- e.g.:- Freshers

* Don done done :-

- White box testing is done, Code is stable, then it can be given to testing.
- Black Box testing is done, application is working according to requirement and it can be given to customer.
- Acceptance testing is done and software is working according to business requirement, Now it can be moved towards production.
- This process is known as "done done done".

* Retrospet Meeting :-

- It is meeting, where all team members comes together and discuss about the achievement and mistakes

done during the sprint of an software.

• This retrospect meetings can be done by :-

- 1> Testing Team.
- 2> Development Team.
- 3> Scrum Team.
- 4> stake holders.

• The outcome of this type of meetings gives the proper way to design and progress the work according to that retrospects backlog in future planning of sprint.